Materials, stress & strain

National 5 · Mechanisms & structures · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…
- ☐ choose and justify a material from a data table
- ☐ calculate direct stress and strain

I DO — Worked example

Read how a full-mark answer is built

A steel rod of diameter 10 mm carries a tensile load of 5 kN.

1 Calculate the stress in the rod. (3 marks)

Full-mark answer each ✓ is one mark

$A = \pi \times 5^2 = 78.5$ mm² ✓. $\sigma = F \div A = 5000 \div 78.5$ ✓ $= 64$ N/mm² ✓.

Why marks are not awarded

"σ = 5 ÷ 78.5 = 0.064 N/mm²"

2 marks: kN not converted to N.

WE DO — Guided practice

Use the starters and word bank to help you

A 2 m wire stretches by 1.2 mm.

2 Calculate the strain. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which justification scores?

☐ It is strong, so it is a good choice for the boat ☐ It is cheap and easy for the builder to get hold of ☐ It is a good material that will last a long time ☐ It is corrosion resistant, so it won't degrade in sea water

b) Strain has units of…

☐ N/mm² ☐ mm ☐ None ☐ N

c) A 1152 N force on 36 mm² gives a stress of…

☐ 41 472 N/mm² ☐ 32 N/mm² ☐ 0.031 N/mm² ☐ 1188 N/mm²

4 Key terms: write the letter of the matching description next to each term.

1. **Stress**	____	**A** Name the property and link it to the job.
2. **Strain**	____	**B** $\sigma = F \div A$ (N/mm²)
3. **Justifying a material**	____	**C** $\varepsilon = \Delta l \div l$ (no units)

A hydrofoil mast must be strong, light and resist corrosion in seawater.

5 Choose a suitable material from aluminium, mild steel and carbon fibre, and justify your choice. (2 marks)

During a test, a steel rod stretches 0.6 mm. The strain was 0.0012.

6 Calculate the original length of the rod. (2 marks)

A steel wire must support a load of 8.0 kN. The stress must not exceed 200 N/mm².

7 Calculate the minimum diameter of the wire. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?